# Week 13 web page: delivery guide

Instructor notes for Week13_HierarchicalModelsAndGAMs.html

## General

- Page: https://kevintshoemaker.github.io/NRES-746/Week13_HierarchicalModelsAndGAMs.html. Script: `CODE_Week13_HierarchicalModelsAndGAMs.R` (same seeds, same numbers; each H1's section runs on its own after the first few lines).
- Monday, Nov. 16: *Formula to Model* review (due that day), the *Partial Pooling by Hand* WOD, then Bayesian hierarchical models (moved from the Week 12 page, which keeps a pointer).
- Tuesday, Nov. 17: final-project check-ins (30 min per group). No lab lecture, Act, or key.
- Wednesday, Nov. 18: splines, GAMs, and Gaussian processes. No in-class WOD; hand out the take-home *Spline by Hand* (due Mon., Nov. 23) at the end. Keys for both WODs: `wod/answer_keys/`.
- Thanksgiving: no class Wed., Nov. 25, so spatial models get one lecture and Lab 10. The GP section is the bridge; if it isn't reached Wednesday, open Nov. 23 with it. Going over *Spline by Hand* on Nov. 23 (~5 min) competes with that lecture; consider doing it during the Lab 10 opening instead.
- Data, all simulated (say so):
  - Drought sensitivity (seed 12), unchanged from Week 12: 12 stands, 194 stand-years; truth $\mu_a = 0.9$, $\mu_b = 0.25$, $\tau_a = 0.15$, $\tau_b = 0.10$, $\rho = -0.6$, $\sigma = 0.12$.
  - Bd thermal gradient plate (seed 13): 48 cultures, 4-29 °C; truth a two-piece curve (Gaussian rise, quadratic fall), peak 0.30/day at 20 °C, zero at 28 °C, noise SD 0.03.
  - Take-home: reservoir chlorophyll, `data/ReservoirChlorophyll.csv` (44 of 52 weeks; spring bloom near week 15, small fall bloom near week 40). Push it to `gh-pages` and check the URL before handing out the WOD.
- Questions on the page have no answers. *Land:* is where discussion should end up.

## Part 1: Bayesian hierarchical models (Monday, ~50 min)

**Before class:** remind students (Canvas) to bring a laptop with `cmdstanr`, `brms`, and `lme4` installed. In class they run the first H1 of the script from the top (about 2 min including compiles).

**Formula to Model review (~8 min)**

- Go over the count table (Part 1a) and one formula-to-equations item. Key in `wod/answer_keys/`.

**WOD: Partial Pooling by Hand (~10 min real; 5 alone, then neighbor)**

- $w_A = 0.5$, $w_B = 0.8$; $\hat{a}_A = 3.6$, $\hat{a}_B = 3.8$. The ranking flips: pond A's raw mean is higher (4.2 vs. 4.0), but 4 frogs give a sampling variance (0.25) as big as the among-pond variance (0.25).
- One-line board summary: $w_j$ = signal variance / (signal + sampling variance). This replaces the formula derivation in section 3.

**Random effects as priors (~2 min)**

- Hierarchical = multilevel = mixed. Four bullets: no integral; uncertainty carries through; priors help with few groups (the singular four-stand fit from Week 12 Q3); the price.

**Partial pooling (~6 min)**

- Code check: the formula matches `lmer()` to three decimals. $\hat{\sigma}_b = 0.113$, $\hat{\sigma} = 0.229$, $\hat{\mu} = 0.861$; weights 0.88 (30 yr) down to 0.55 (5 yr).
- Widget (`partial_pooling.html`): $\sigma_b = 0$ (every stand on the dashed line; weights 0.00), then $\sigma_b = 0.4$ (weights 0.99 and 0.94, nearly no pooling).
- "Set $\sigma_b$ to 0, then drag it to 0.4. Which two pooling choices are these? Then compare S10 and S12 ... Which moves more toward the overall mean, and why?" *Land:* complete pooling, then (nearly) no pooling. S12 (5 yr, $w = 0.55$) moves a larger share of its distance (45% vs. 34% for S10, $w = 0.66$); absolute moves are almost equal (0.080 vs. 0.078 mm) because S10 starts farther out.

**Priors for a covariance matrix (~4 min; cut to 1 min if behind)**

- Board: $G = \text{diag}(\tau)\,\Omega\,\text{diag}(\tau)$; LKJ gives $p(\rho) \propto (1 - \rho^2)^{\eta-1}$ for $2 \times 2$.
- Widget (`lkj_prior.html`), $2 \times 2$: $\eta = 1$ flat ($P(|\rho| > 0.5) = 0.50$); $\eta = 2$ 0.31; $\eta = 0.5$ U-shaped, 0.67. Ask for a prediction before switching to $10 \times 10$.
- "For a $10 \times 10$ correlation matrix with $\eta = 1$, how likely is any one correlation to exceed 0.5 in absolute value? What does 'uniform' mean for a matrix that large?" *Land:* about 0.10 (prior SD 0.30). LKJ(1) is uniform over valid matrices, and most valid big matrices have small correlations.

**Stan walkthrough (~13 min; laptops)**

- Two ideas: the Cholesky factor (always a valid $\Omega$) and non-centering (the Week 9 funnel). `ab[stand, 1]` is multi-indexing.
- Students compile and sample. No divergences; all $\hat{R} = 1.00$.
- Table: everything matches `lmer()` except $\rho$: MLE $-0.57$; posterior median $-0.37$, 90% interval $-0.76$ to 0.17; truth $-0.6$.
- Shrinkage figure: S11 (5 yr) moves most; S01 and S02 (30 yr) barely move. Truth

check: in this data set the short-record slopes barely improve (0.057 to 0.055); over 500 simulations slope error falls 0.070 to 0.051. Shrinkage wins on average, not every time.
- "Why is the posterior for $\rho$ so much wider ... closer to 0 than the MLE? Which summary is more honest?" *Land:* 12 stand-level pairs, each estimated with error; the LKJ(2) prior pulls toward 0 and integrating over the $\tau$s spreads it. The posterior is more honest.
- "Change to `lkj_corr_cholesky(1)`, then `(10)` ..." *Land:* $\eta = 1$: median $-0.45$ (90% $-0.81$ to 0.12); $\eta = 10$: $-0.13$ ($-0.45$ to 0.21). Stand slopes move by at most 0.008 and 0.026. Split the room.

**brms (~5 min)**

- Same model and priors: cor $-0.33$ (95% $-0.81$ to 0.30), SDs 0.16 and 0.11. Have students find `lkj_corr_cholesky`, `diag_pre_multiply`, and `z_1` in `stancode(fit_brms)`.

**Discussion (if time; otherwise assign Q3 for project check-ins)**

- Q1 (12 more stands or 20 more years?) *Land:* more stands; $\rho$ and the $\tau$s describe the population of stands, and longer records can't add stands.
- Q2 (brms vs. your own Stan) *Land:* brms for anything a formula can say; your own Stan for custom process or observation models.
- Q3 (your project's groups) *Land:* open; ask for the grouping factor, its number of levels, and whether slopes plausibly vary. Good prep for Tuesday's check-ins.

**Skip if behind:** the LKJ widget (state the prior in one sentence), the $\eta$ experiment, the discussion. The ridge-as-prior bonus is now needed Wednesday; mention it in one sentence at the end.

# Part 2: Splines, GAMs, and Gaussian processes (Wednesday, ~50 min)

No laptops needed; all code is on the page.

**Big picture and the bell curve (~6 min)**

- MODELS box: when no bestiary function has the right shape, build one. Management question: above what temperature does Bd stop growing?
- Bell fit: $r_{\max} = 0.314$, $T_{\text{opt}} = 19.4$, width 4.72, $\sigma = 0.038$.
- "Which features of the scatter does the bell curve get wrong? Which matter for the management question?" *Land:* symmetric (the drop above the optimum is steeper than the rise), can't go below 0 (cultures die above ~28 °C), overpredicts at 27-29 °C. The upper tail is the one the question is about.

**Splines are linear models (~12 min)**

- Board: $f(x) = \sum \beta_k B_k(x)$; each $B_k$ is a column of $X$. Show `head(bs(...))`: most

entries 0 (local bumps). Knots at 10.4, 16.4, 22.65 °C for 6 functions.

- Figure: 6 functions fit well; 25 chase single cultures. Callback to the Week 10 connect-the-dots WOD.
- Widget (`spline_basis.html`), stops in order:
  1. B-spline, degree 1, 4 knots, "show each weighted basis function": tents; the sum bends only at knots. Ask where it can bend first.
  2. Degree 3: smooth joins (the "Where pieces join" card).
  3. Polynomial, highest power 12, randomize 3-4 times: most of the motion is at the edges (powers blow up near $\pm 1$).
  4. Back to B-spline, 12 knots, independent weights: jagged. Switch to random walk and drag the step SD from 1 down to 0.05: smooth.
- Q1 (degree 1 vs. 3; polynomial) *Land:* degree 1 bends only at knots (kinks); cubic joins are smooth. Polynomials move everywhere when one weight changes and swing hardest at the ends.
- Q2 (random walk) *Land:* smooth curves come from weights that change slowly from neighbor to neighbor; that assumption is the penalty.

**Penalizing wiggliness (~12 min)**

- Board: penalized NLL with second differences; $\lambda = 0$ is `lm()`, $\lambda \to \infty$ is a straight line; edf from $K$ down to 2.
- Penalty = prior: $\lambda = 1/(2\tau^2)$, so the wiggly part is a random effect and REML picks $\lambda$ the way `lmer()` picks $\sigma_b$. Callback to Monday's ridge bonus.
- Widget (`penalized_fit.html`), stops: $K = 40$, $\log_{10}\lambda = -4$ (edf 38.9, wiggly); drag up to 5 (edf 2.06, a sloped line). "Jump to the REML minimum": $\log_{10}\lambda = 0.9$, edf 9.1. Then $K = 20$ (minimum at 0.0, edf 8.1) and $K = 10$ ($-1.2$, edf 6.7). Tick "show the curve the data were simulated from" last.
- Q (edf and the sloped line; how much $K$ matters) *Land:* edf falls to 2; the second-difference penalty is zero for any straight line, so the limit is the least-squares line, which slopes up (0.006 per °C). Once REML picks $\lambda$, $K = 10$, 20, 40 give edf 6.7-9.1 and nearly the same curve: $K$ only needs to be big enough.
- `gam()`: edf 8.2; `gam.vcomp()` prints the smooth's SD (0.017). Default GCV picks a wigglier curve (edf on the page); always ask for REML.
- AIC table: bell $-170.6$, 6-function `lm()` $-199.5$, GAM $-201.2$ ($\Delta$AIC 30.6 for the bell). GAM peak 20.2 °C, zero at 28.0 °C (truth 20 and 28); bell peak 19.4, never reaches 0.

**Gaussian processes (~10 min; first thing to cut, moves to Nov. 23)**

- Board: MVN on $f$ itself; kernel $\alpha^2 \exp(-d^2/2\ell^2)$. Exponential kernel = AR(1) with $\rho = e^{-1/\ell}$ (point at the Week 12 $V$ heatmap).
- Fit by `optim()`: $m = -0.057$, $\alpha = 0.32$, $\ell = 10.1$ °C, $\sigma = 0.027$. GP and GAM curves differ by at most 0.011. Don't dwell on $m$ (it trades off with $\alpha$ when $\ell$ is long).
- Widget (`gp_draws.html`), stops: no points, squared exponential, $\ell = 1.5$ (correlation 0.80 at distance 1; falls to 0.1 at 3.2); "New random functions" twice. Add example points. Switch to exponential (0.51 at distance 1): rough draws, posterior band pinches at each point. Back to squared exponential, $\ell = 0.3$: between and beyond the points

the posterior returns to the prior mean (0) and the band widens. Ask for a prediction before each switch.

- Q *Land:* exponential curves are continuous but rough (like an AR(1) series); squared exponential ones are smooth. With a short $\ell$ the data only inform nearby $x$; far away you get the prior back.
- Close: replace temperature with distance between sites, and this is a spatial model (kriging).

**Discussion (~5 min; pick one)**

- Q1 (32 °C: bell 0.009, GAM −0.315, GP −0.265) *Land:* all three are extrapolating. The bell can't go negative; the GAM and GP continue the last slope. None should be used; measure at 30-34 °C.
- Q2 (GAM beats bell by 30.6) *Land:* the bell is falsified as a description; asymmetry is corroborated only against this one rival. Add an asymmetric parametric rival (Briere, Sharpe-Schoolfield, or the two-piece Deutsch et al. 2008 form that generated the data).
- Q3 (manager wants one number) *Land:* report $\mathrm{CT}_{\max}$ from the GAM with a simulation interval (bonus: median 28.0, 95% 27.7-28.4; $T_{\mathrm{opt}}$ 20.2, 18.5-21.6), or fit an asymmetric parametric curve with $\mathrm{CT}_{\max}$ as a parameter.

**Last 2 min:** hand out *Spline by Hand* (due Mon., Nov. 23). Take-home, laptops, AI labels on each coding item. Numbers they should get: AIC −5.9, −16.4, −41.8, −39.6 for $K = 4$, 8, 12, 20; `pnll` check 43.72; $\lambda = 10$ is closest to `gam()` (edf 9.7, peak week 15.5). Don't show the penalized NLL code before it's due; the page uses `gam()` only.

**Bonuses (not planned for class):** derived-quantity intervals; the `brms` smooth (`sds(stemp_1)` 0.53, on a different scale from `gam.vcomp()`); a table of GAM formula forms for final projects (Pedersen et al. 2019).